TEZPUR UNIVERSITY TU/CSE

Autumn Mid Semester Examination, 2025

CSMT505: RESEARCH METHODOLOGY

Full Marks: 30 *Time: 90 min*

The figures in the right-hand margin indicate marks for the individual question

*(Answer any **three** questions for 30 marks)*

1. (a) What is simulation? Give example. **2+4+4 = 10**
   (b) Write the seven major steps to be executed in a simulation study.
   (c) Justify its usefulness in experimental research with example.

2. (a) Write the major components of a survey article. **3+(2+2)+(1+2)=10**
   (b) When do you say a system or method is scalable? How do you differentiate 'load scalability' from 'space scalability'?
   (c) What is a patent? What are the major steps for submitting a patent application?

3. Write brief notes (*any five*): **2x5=10**
   a. *h*-index
   b. Impact Factor
   c. SCI journal
   d. Parametric analysis
   e. Non-parametric analysis
   f. Copyright

4. (a) What is text plagiarism? Give an example. **1+1+2+3+3=10**
   (b) What is code plagiarism? Give an example.
   (c) What is paraphrasing? Give an example.
   (d) What measures will you take to avoid text plagiarism? Explain.
   (e) What will be the consequences if anyone found guilty of plagiarizing an original author's work?

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CSMT505 Research Methodology

Time: 30 mins. **Marks: 10**

1. State *true* or *false* (any *twenty*) 0.5x20=10

a) To solve a research problem, we need to be systematic and planned.
b) An MTech dissertation needs to include some significant work in a defined field of research.
c) MTech dissertation research is not time bound.
d) MTech dissertation problem can be both theoretical as well as experimental.
e) Every MTech dissertation research is based on some research question(s).
f) Problem finding and formulation is not integral to research.
g) The goodness of a research question can be defined in terms of its previous solutions, do-ability and usefulness in the defined field of research.
h) Publication of paper based on dissertation content is helpful for visibility.
i) After selection of the research question, the student must be enthusiastic about it.
j) Literature survey helps improve breadth of knowledge.
k) Re-doing bad work can also help contribute to experimental research.
l) Facing criticism is not helpful in deciding appropriate track for quality research.
m) Simplification of any problem is not possible.
n) Exhaustive experimentation is enough for successful completion of any experimental research.
o) Single-minded dedication may affect quality or completion of a research work.
p) Abstract and introduction are same in a manuscript.
q) Discussion of tabulated results is essential in a manuscript.
r) Presenting partially cooked ideas helps improve the quality of work.
s) Criticism can be valid as well as invalid.
t) Attending seminars and conferences clarify a research problem.
u) To improve quality of research, the inclusion of theorems and lemmas may be useful.
v) Spending more time in a laboratory may not be helpful if concepts are not clear.

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TEZPUR UNIVERSITY **TU/CSE**
Autumn End Semester Examination, 2025
CSMT505: RESEARCH METHODOLOGY

Full Marks: 50 *Time: 120 min*

The figures in the right-hand margin indicate marks for the individual question
*(Answer **Section B** and any **three** questions from **Section A**)*

SECTION A

1. (a) How do you ensure quality in data? **2+8=10**
(b) Why are the following characteristics important for quality data?
(i) Completeness, (ii) Correctness, (iii) Relevance, (iv) Recency

2. (a) State the importance of the following statistical measures in data analysis: **6**
(i) True Negative (ii) False Positive (iii) Precision (iv) Accuracy
(b) State the utility of χ^2 test in data analysis. **4**

3. (a) Do you think Generality testing is important? If so, why? **4**
(b) What are internal and external generality tests? Give examples. **4**
(c) What are two basic limitations of generality test? **2**

4. (a) State the use of the following sampling techniques: **3**
(i) Random (ii) Stratified (iii) Systematic
(b) Give a pseudocode to eliminate redundant (or duplicate) instances in a dataset. **4**
(c) Describe an approach to select relevant features from a dataset. **3**

SECTION B

5 Write the code snippet for the following algorithm, use necessary packages : **3**

$i \leftarrow 10$;
if $i \geq 5$ **then**
  $i \leftarrow i - 1$;
else
  if $i \leq 3$ **then**
    $i \leftarrow i + 2$;
  end
end

Algorithm 1: Algo 1

6. What are the functionalities of the following (*any four*): **4**
\newtheorem{theorem}{Theorem}, block, \mode<presentation>, wrapfig, hyperref, cite

7. For the following table write the code for generating it in LaTeX . **5**

Table1: Details of students

Student name	Department	CGPA
Ahan	MBBT	8
Ruby	Chemistry	--
Noah	Physics	8.5

8. Write the code snippet for inserting figures in LaTeX using examples and packages. **5**

9. What are the three types of blocks in beamer, explain with examples (write the LaTeX code for them and output produced). **3**

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